LANGUAGE AS A CLINICAL INTERFACE: NLP-BASED ASSESSMENT, ILLNESS MONITORING, AND RESPONSIBLE AI IN SCHIZOPHRENIA

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16th Mind-Body Interface International Symposium · Symposium 24 · Taichung, 15 November 2026

MBI 2026
C O I Disclosure

Presenter : Hironobu Nakamura

There are no companies, etc. in a relation of conflict of interest requiring disclosure by the presenter(s) in relation to the contents of the presentation.

The name of the ethics review organization: Medical Research Institute Tokyo Medical and Dental University.
The approval number: M2020-057

WHY I MEASURE LANGUAGE

About me

Psychiatrist and epileptologist; Assistant Professor and Chief Ward Physician, Dept. of Psychiatry, Institute of Science Tokyo Hospital

Ph.D. (2024), Tokyo Medical and Dental University

UNDERPIN project (Keio University, Prof. Kishimoto): NLP of psychiatric speech

Awards: Association for Natural Language Processing (2024); Japanese Society for Artificial Intelligence (2023)

“Language is where psychopathology becomes observable.”

Words are our richest clinical data, yet for a century we could only describe them. NLP lets us measure them.

Today

1 History

2 Methods

3 Our study

4 The map

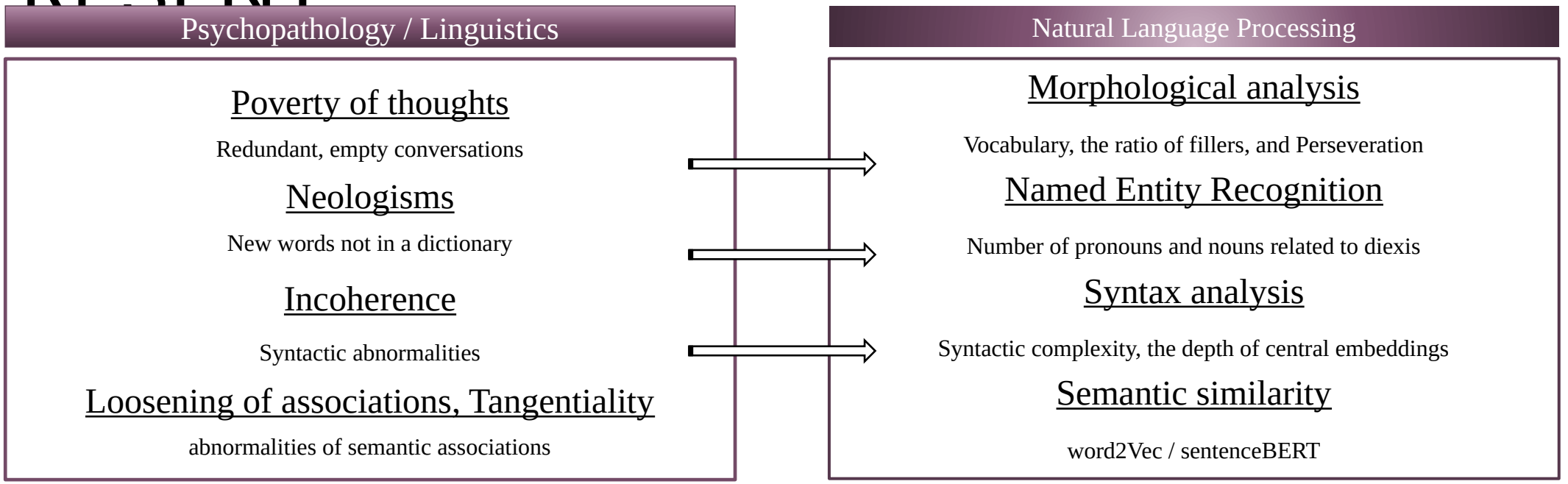
5 Promise & peril

TWO HISTORIES CONVERGE

	1900s–1960s	1970s–1990s	2000s–2012	2013–2019	2020–now
Psychiatry & psycho-pathology	Bleuler (1911): loosening of associations	Andreasen (1979): TLC scale, clinician-rated thought disorder	LSA coherence in schizophrenia (Elvevåg 2007); speech graphs (Mota 2012)	Speech predicts psychosis onset (Bedi 2015; Corcoran 2018)	Multilevel, multilingual, longitudinal; UNDERPIN (2022); our study (2026)
			Convergence: computational psychiatry of language		
Natural language processing	Rule-based systems; ELIZA (1966)	Statistical NLP; latent semantic analysis (1990)	Machine learning; parsers and corpora	Word2Vec (2013), Transformer (2017), BERT (2018), SentenceBERT (2019)	Large language models: GPT-3 (2020), ChatGPT (2022); multimodal, agentic AI

CONCEPTUAL DIAGRAM : PAST AND PRESENT

From Classical Psychopathology to NLP



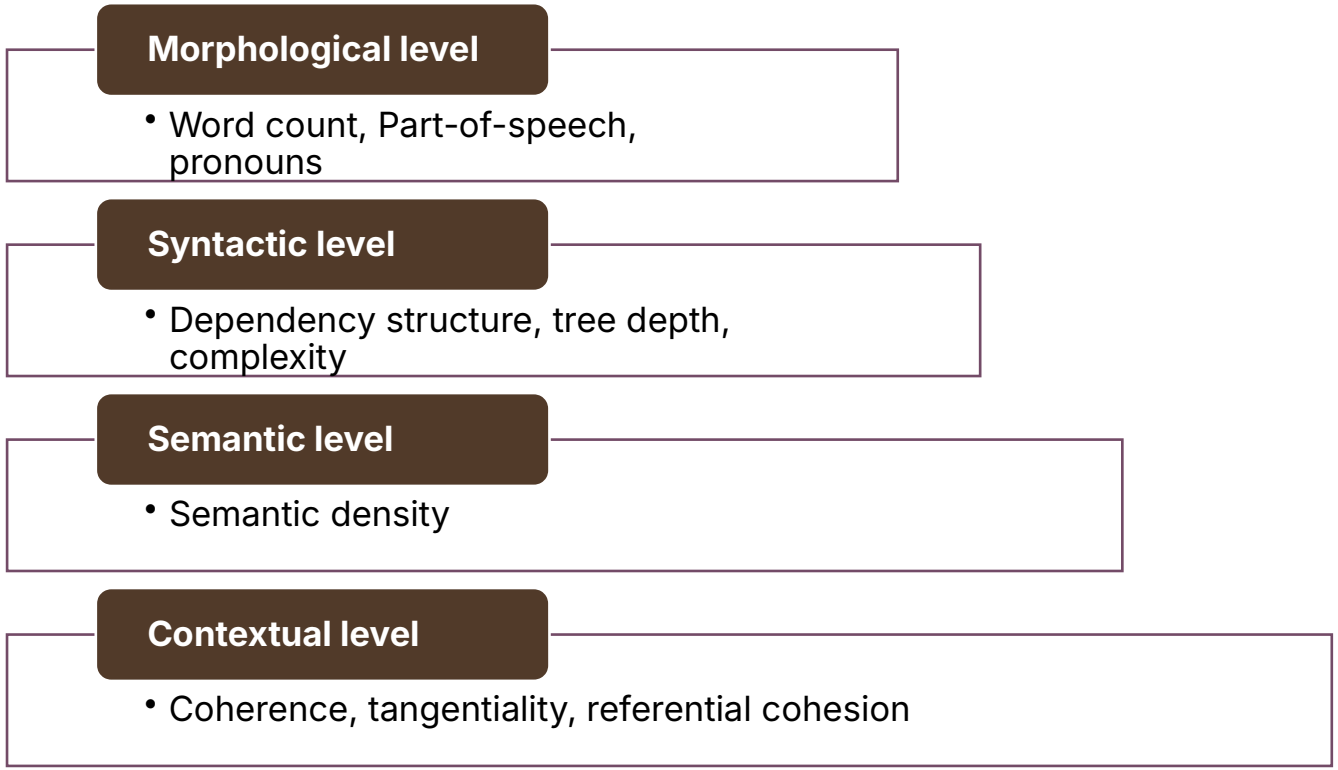
- Transition: from descriptive interpretation → integrative, quantitative NLP
- Goal: bridge classical insights with modern computational tools, combining hypothesis-driven and data-driven approaches (a hybrid).

INTRODUCTION

- **Language in schizophrenia** has long been studied.
- **Natural Language Processing (NLP)** enables quantitative analysis of linguistic abnormalities.
  - Clinical applications: detection, monitoring, prediction (onset, prognosis, relapse).
- **Findings so far:**
  - ↓ **Syntactic complexity**
  - ↓ **Semantic coherence**
  - Other features: poverty of content, cohesion, metaphor, emotion, pronouns.
- **Computational models:**
  - *Speech graph* (positive FTD)
  - *Semantic density* (negative FTD)
  - *Vector-based models* (LSA, coherence).
- **Limitations:**
  - Narrow focus, small samples, single-task designs.
- **Our study:**
  - Large-scale, multi-feature NLP analysis
  - Hypothesis: **reduced syntactic complexity & increased semantic proximity** as core abnormalities.

CONCEPTUAL DIAGRAM 1: INTEGRATIVE APPROACH

Levels of Linguistic Analysis in NLP for Schizophrenia



- Prior studies often used these *in isolation*.
- **Our approach**: integrative, comprehensive, large-sample

FOUR FAMILIES OF NLP MODELS

Rules & counts	Static embeddings	Contextual embeddings	Large language models
Examples Dictionaries (LIWC), POS tagging & parsing (GiNZA), TF-IDF	**Examples** Word2Vec, GloVe	**Examples** BERT, SentenceBERT	**Examples** GPT-class and other foundation models
Captures Which words and structures are used	**Captures** How close words are in meaning	**Captures** Meaning in context; coherence between sentences	**Captures** Zero-shot rating, summarising, generating text
Interpretability **High**	**Interpretability** **High–medium**	**Interpretability** **Medium**	**Interpretability** **Low (black box)**

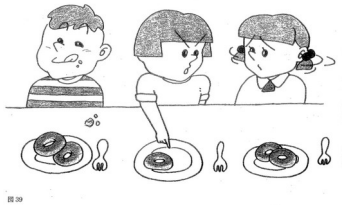
More interpretable → **More representational power**

Our study combined the first three families so that every marker can be explained to a clinician.

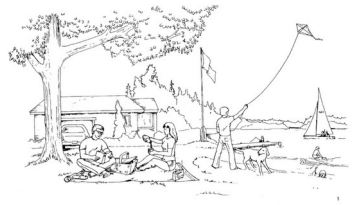
MATERIAL & METHOD: PARTICIPANTS

Nakamura H, et al. Psychological Medicine 2026;56:e193. doi:10.1017/S0033291726104668

- **Sample**
  - 104 patients with schizophrenia (diagnosed by DSM-5 or ICD-10), and 101 healthy controls
  - Age ≥ 20 years
  - All fluent in Japanese
- **Interview sessions** (30–60 min)
  - **Free conversation**: illness, daily life, hobbies, news
  - **Picture description**: standardized images (VPTA, COGNISTAT, WAB)
  - **Storytelling**: *Cinderella* or *Kaguya-Hime*
- **Study framework**
  - UNDERPIN project; 7 hospitals + 3 outpatient clinics in Japan; 2018–2024
  - 2–5 visits per participant; PANSS for symptom severity
- **Ethics**
  - Written informed consent; approved by the IRBs of Keio University and participating institutions


(a)


(b)


(c)

METHOD: SPEECH ANALYSES

- **Transcription & Corpus**
  - All speech data were transcribed into text.
  - Only **patients' speech** was analyzed (the interviewer was excluded).
- **Feature Extraction**
  - **Morphology:**
    - Part-of-speech tagging, and Named Entity Recognition related to time, place, person, and god (GiNZA).
  - **Syntax**
    - Dependency distance, Tree depth, Depth of central embeddings, and Other measures of syntactic complexity
  - **Semantics**:
    - Word similarity (Word2Vec) , Semantic network analysis (Word2Vec + TF-IDF)
  - **Contextual level**:
    - Sentence similarity (cosine distance, sentenceBERT), and Perseveration

Feature Processing	76 NLP features per task & visit Ratio-defined(length adjusted), z-standardized(train set parameters)
Exploratory Analyses	Group comparisons (SCZ vs HC, **univariate GEEs)** Results descriptive only (not used for final selection)
Dimensionality Reduction	Exploratory factor analysis Representative feature = largest absolute loading (adjusted for correlations/multicollinearity)
Validation (Training set)	Task-specific **GEEs** (Outcome: SCZ (1) vs HC (0)) Predictors: Representative features + age, sex
Predictive Evaluation (Test set)	AUC, sensitivity, specificity All preprocessing & feature selection confined to training set

METHODS: ANALYSIS PIPELINE

RESULTS: DEMOGRAPHICS

	SCZ	HC	difference
Sample size	104	101	N/A
Age (years)	48.3 ± 12.7	51.3 ± 18.1	p = 0.166
Gender (female, %)	48.1	55.4	p = 0.360
PANSS total	63.3 ± 18.0	30.7 ± 1.3	p < 0.001
PANSS positive	15.4 ± 6.0	7.2 ± 0.5	p < 0.001
PANSS negative	16.4 ± 6.3	7.3 ± 0.4	p < 0.001
PANSS general psychopathology	31.4 ± 8.6	16.3 ± 0.7	p < 0.001
Task participation - Free conversation	104	101	-
Task participation - Picture description	102	101	-
Task participation - Storytelling	101	100	-
years of education (mean ± SD) (Available N)	13.3 ± 2.4 (N = 51)	15.4 ± 2.8 (N=70)	p < 0.001

SCZ = schizophrenia. HC = healthy controls.

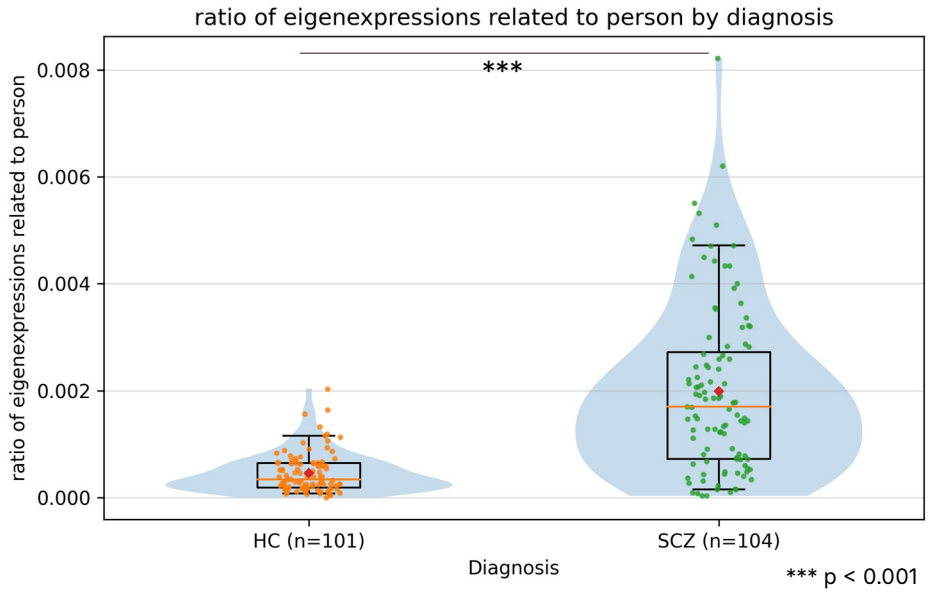
METHODS: ANALYSIS PIPELINE

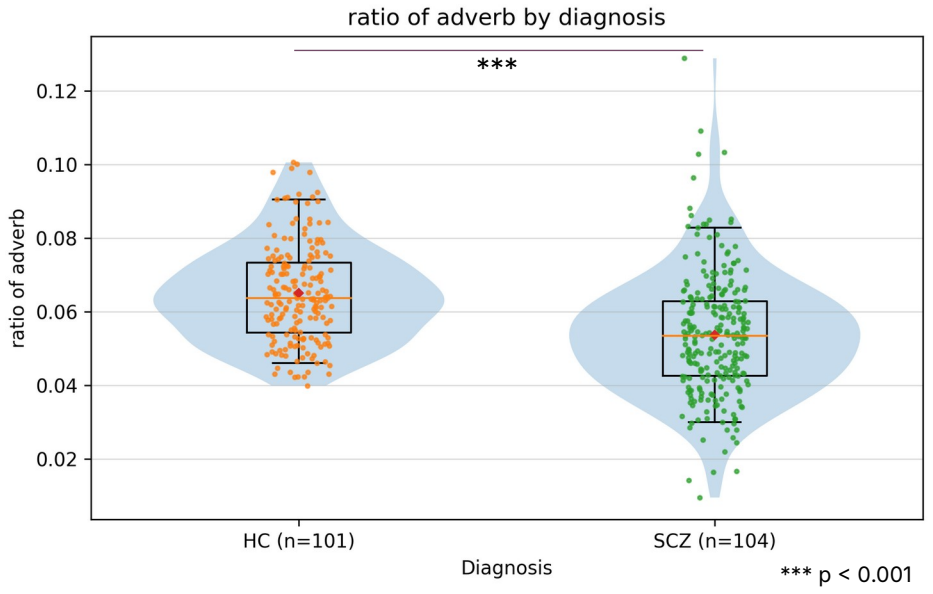
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RESULTS: EXPLORATORY ANALYSES

Group differences in Task 1: Free conversation

- Group Comparison (SCZ vs HC, **univariate GEE**): **35**/76 features significant (FDR-corrected)

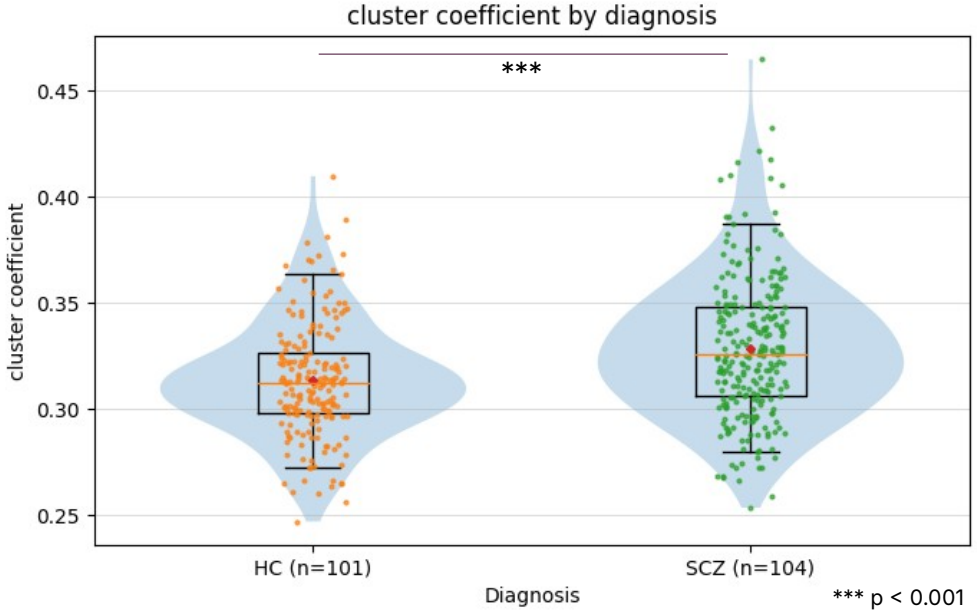


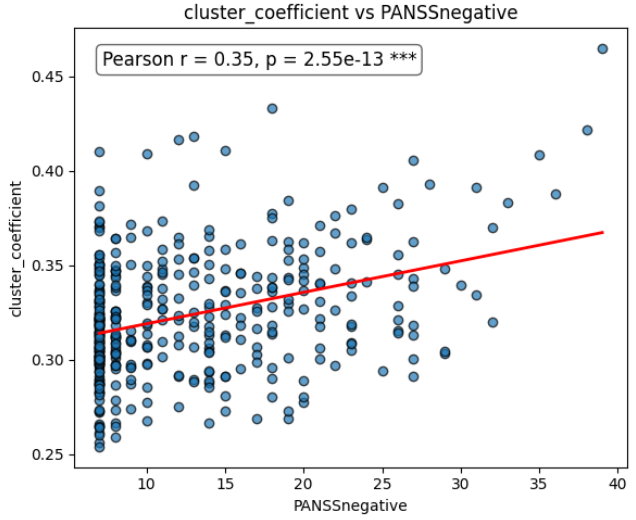


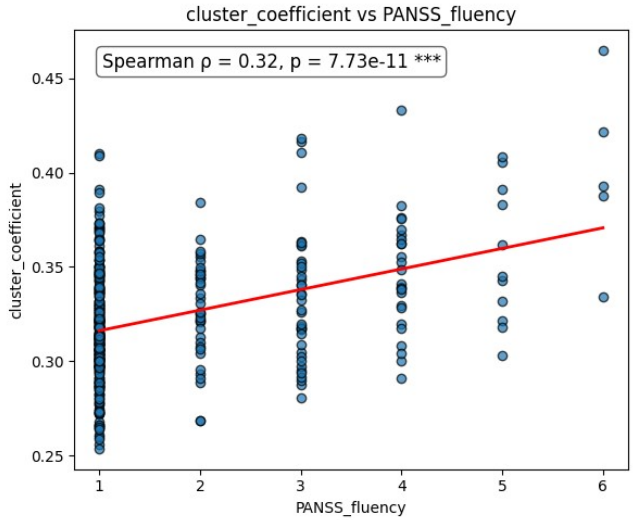
RESULTS: EXPLORATORY ANALYSES

Group differences in Task 1: Free conversation

- The “cluster coefficient”, a semantic network feature, also reveals group differences and is correlated with the PANSS negative score and the Fluency of speech (PANSS

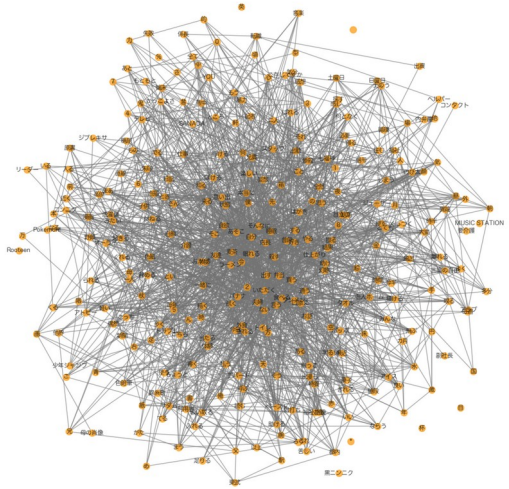




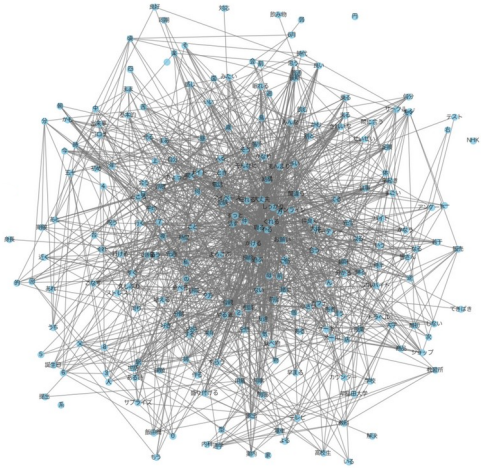


Semantic Network Analysis Using a TF-IDF Framework

- Using the comprehensive UNDERPIN dataset, develop a Term Frequency–Inverse Document Frequency (TF-IDF) model, which calculates a measure of frequency and importance of a word in a document.
- Extract the top 10% high-score (TF-IDF score) words from the subject's speech.
- Form graphs (Fig. 1) with content words as nodes; connect N=10 words exceeding the sth = 0.3 similarity threshold until fewer than ***N=10*** remain (Utilize NetworkX in Python for graph creation).
- Calculate the following indicators using network analysis methods: the cluster coefficient, the diameter, the average distance, and the average closeness centrality.



A subject with severe SCZ



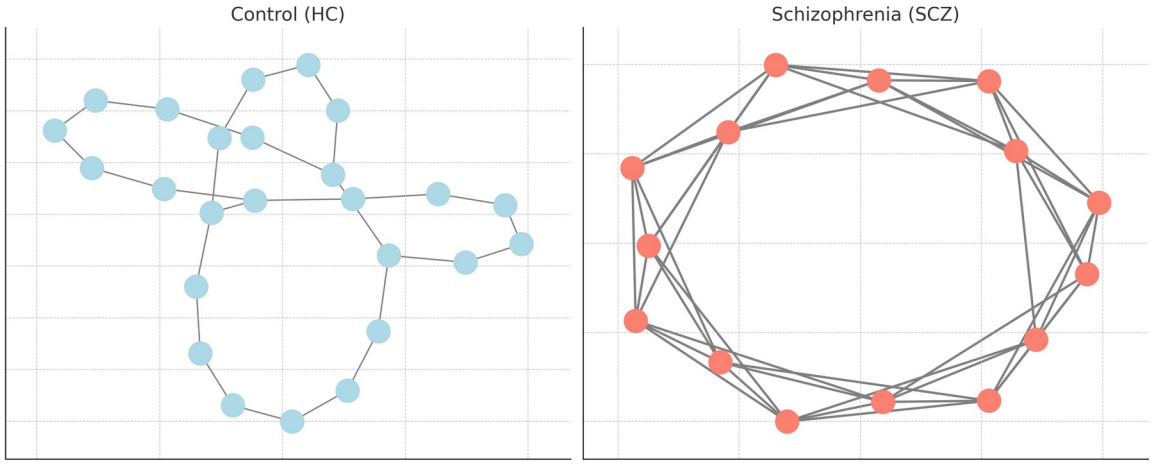
A healthy control

Fig 1. Examples of Semantic Graphs

SUMMARY OF RESULTS: SEMANTIC NETWORK ANALYSIS

Group differences in Task 1: Free conversation

- In schizophrenia
  - ↑ increase: **Cluster coefficient *, density ***
  - ↓ decrease: **average distance *,** diameter, average closeness centrality



Semantic Network in SCZ:
Semantic connections are too tightly linked both locally and globally, but this over-connectedness does not enhance information transfer efficiency.

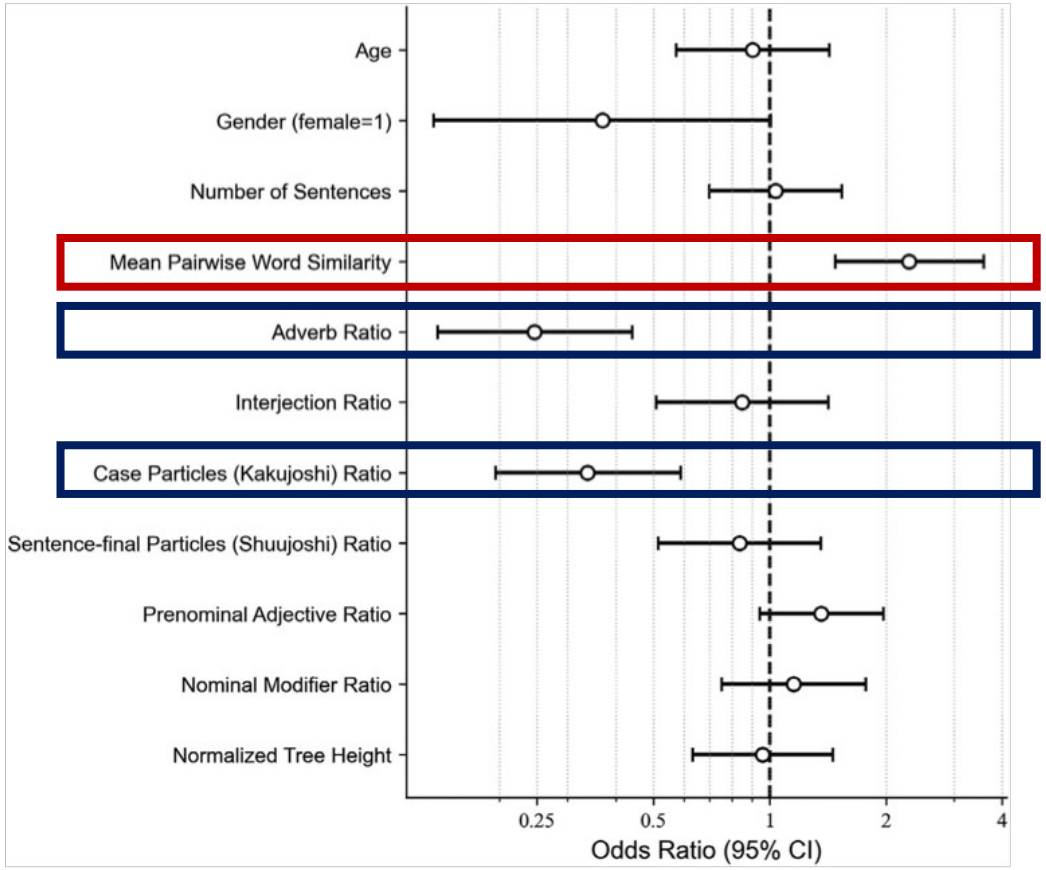
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RESULT: FACTOR ANALYSIS & GEE (TASK 1)

Robust Factor Structure and Diagnostic Performance

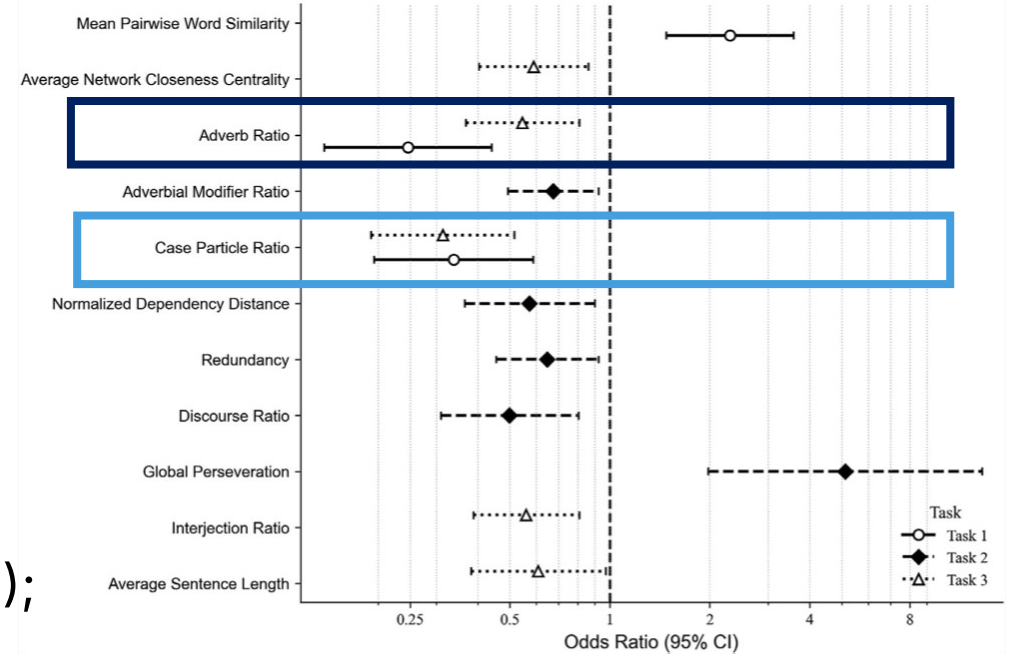
- **Nine-factor solution** identified; nine representative features retained (all VIF < 3).
- SCZ was associated with **higher mean pairwise word similarity** and **lower case-particle and adverb ratios**.
- GEE model showed strong independent-set discrimination: **AUC 0.87** (95% CI, 0.74–0.97; permutation $p < 0.001$).



RESULT: COMPARISON OF TASKS

Cross-Task Comparison of Speech Features

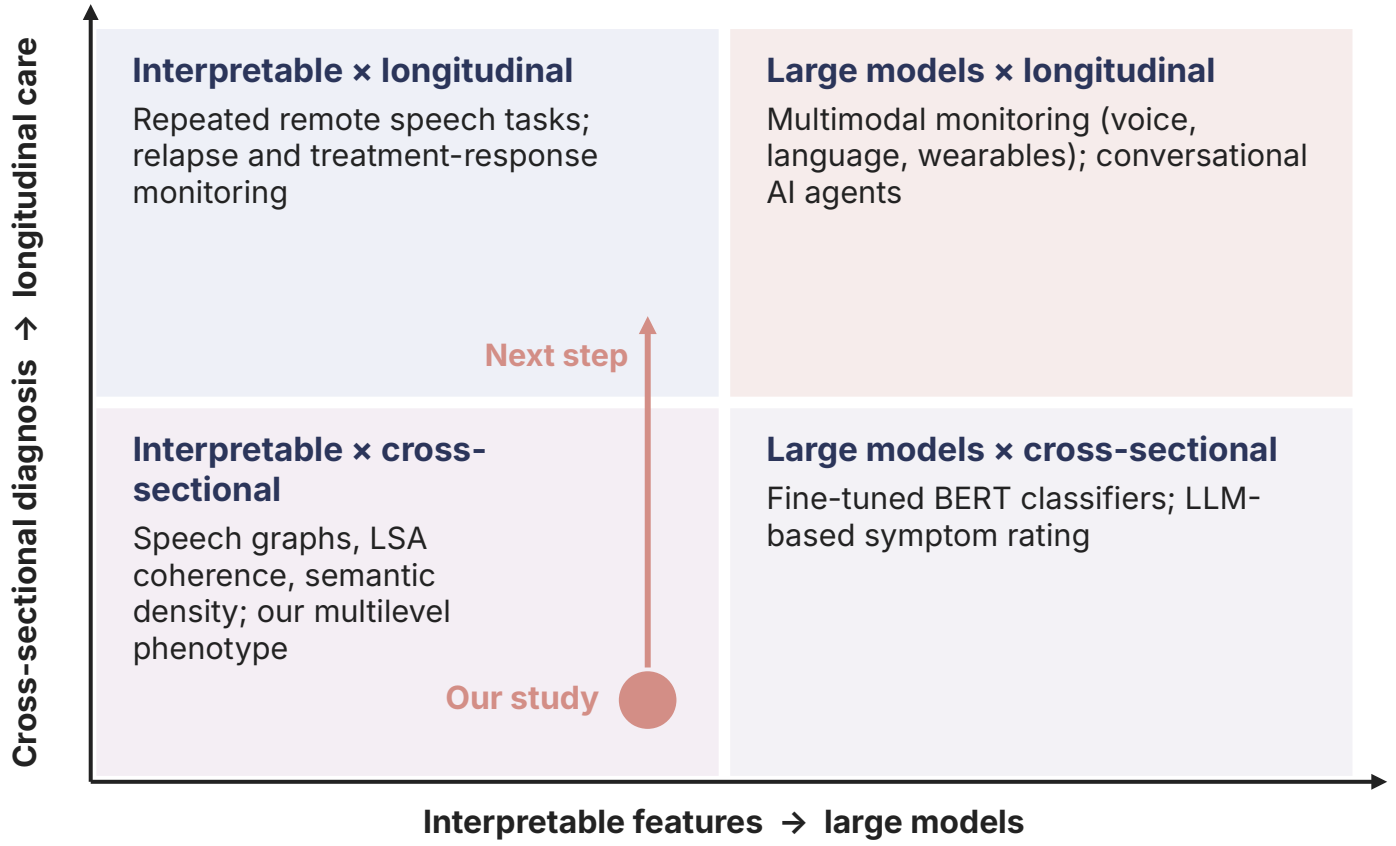
- Adverbial, semantic-network, case-particle, and discourse features showed diagnostic relevance across tasks.
- **Adverb Ratio** was cross-task consistent but task-dependent, with the strongest effect in free conversation (*q*PC = 0.021; interaction $p < 0.001$).
- **Case Particle (Kakujoshi) Ratio** showed robust, task-invariant syntactic simplification (*q*PC = 0.014); age and sex were non-significant.



DISCUSSION

- **Core interpretation:** Our findings suggest a **tripartite linguistic phenotype** of reduced morphosyntactic explicitness, constrained semantic exploration, and weakened contextual modulation. These separable but interacting dimensions extend beyond the positive–negative FTD dichotomy.
- **Language typology:** Japanese's broad adverbial functions and segmentable grammatical markers may increase the detectability of these disturbances. Convergence with Turkish suggests shared vulnerabilities whose expression depends on language typology.
- **Semantic networks:** Locally dense but globally less integrated networks suggest restricted movement across distant semantic domains. This pattern parallels neuroimaging findings (Hayashi et al., 2024; Matsumoto et al., 2023) and may contribute to tangential, perseverative, or impoverished speech.
- **Limitations and future:** Findings characterize established, predominantly mild–moderate SCZ rather than predict onset. Cross-linguistic, longitudinal, and multimodal studies, alongside direct cognitive, functional, and FTD assessments, are needed to test these interpretations.

MAPPING THE FIELD



Five trends

Single feature → multidimensional

English → cross-linguistic

Lab interview → smartphone & remote

Text only → voice, language & wearables

Handcrafted features → large language models

LIMITATIONS: WHAT IS MISSING

- **Sample composition**
  - Only schizophrenia patients (no clinical high-risk group)
- **Severity**
  - Mean PANSS total = 63.3 → mostly mild to moderate cases
  - More severe cases might show additional abnormalities
- **Language & generalizability**
  - Data in Japanese (agglutinative language)
  - Need cross-linguistic validation for broader applicability
- **Speech modality**
  - Written data may better capture semantic coherence

PROMISE AND PERIL

Promise

- Objective, reproducible severity assessment
- Early warning of relapse
- Tracking treatment response
- Continuous monitoring between visits
- Less inter-rater variability; scalable screening

Peril

- Privacy: voice and words identify the person
- Consent, a sense of surveillance, loss of trust
- Bias across languages, dialects, education, culture
- Opacity of large models; hallucination
- Overdiagnosis, labelling, automation bias
- Generative AI echoing or reinforcing delusional content

Language data are intimate data: the signal that informs care can also expose the person.

TOWARD RESPONSIBLE USE

Goal: AI that augments, not replaces, clinical understanding

Evidence

- External and cross-linguistic validation
- Longitudinal, prospective designs
- Clinical high-risk cohorts
- Links to function, cognition, and thought-disorder scales

Ethics & governance

- Privacy by design; de-identification
- Informed, revocable consent
- Bias audits across groups
- Transparency and explainability
- Alignment with medical-device and AI regulation

Clinical integration

- Human-in-the-loop decisions
- Education for clinicians and patients
- Patients and families in tool design
- Judge benefit to care, not AUC alone

CONCLUSION: TAKE-HOME MESSAGES

1

Language is a clinical interface

NLP turns everyday speech into measurable, interpretable markers of psychopathology.

2

A tripartite linguistic phenotype of schizophrenia

Fewer case particles, semantic narrowing, and fewer adverbs: AUC 0.87 in free conversation, consistent across tasks.

3

Responsible AI: augment, not replace

Validation, privacy, fairness, and human responsibility come before clinical use.

Thank you to the UNDERPIN participants and collaborating institutions, Prof. Taishiro Kishimoto, Prof. Hidehiko Takahashi, and Dr. Yoshinobu Kano.